EVQ-Cosh: Closed-Form Frequency Allocation for Long-Context Extrapolation

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Abstract

Long-context language models are typically extended at inference time, but the training-time geometric allocation of RoPE frequencies—a key determinant of long-range representational quality—is usually left untouched. We formulate frequency allocation as a variational inverse problem, derive a broadband surrogate whose stationary density satisfies a closed-form ODE, and obtain a one-parameter family (EVQ-cosh) in which geometric RoPE is the degenerate $\tau{=}0$ limit. In PE-dominant extreme extrapolation (128→8K), EVQ surpasses learnable positional encoding with zero extra parameters (333.7 vs. 455.3 PPL), isolating frequency structure as the decisive variable. More importantly, EVQ provides a better substrate for inference-time scaling: EVQ+YaRN reaches 100% retrieval at 8K (6 seeds) where Geo+YaRN plateaus at 61–65%, and lowers PPL by 14–32% at 8–16K. Progressive context extension (single seed) widens this gap monotonically, reaching 48K functional context (24× training length) at PPL 2.63. On Multi-head Latent Attention (DeepSeek-V2/V3 mechanism), EVQ alone outperforms Geo+YaRN at 2–3× extrapolation (−31.1% PPL, 3 seeds). These results identify training-time frequency allocation as an under-explored but critical design axis for long-context extrapolation.

1 Introduction

Long-context language modeling is often framed as an inference-time scaling problem: if rotary position embedding (RoPE) is compressed, interpolated, or rescaled carefully enough, a model trained at length L_{train} should extrapolate to substantially longer contexts [Chen et al., 2023, Peng et al., 2024, Ding et al., 2024]. This view has produced strong engineering baselines, but it leaves a fundamental design choice untouched: the training-time geometric allocation of RoPE frequencies. Standard RoPE treats that allocation as fixed, even though long-range failure is ultimately mediated by frequency collisions in the learned representation.

We take the opposite starting point. Instead of asking only how to rescale positions at inference time, we ask how frequencies should be allocated during training. We formulate RoPE frequency allocation as a variational inverse problem, derive a broadband surrogate in which the stationary density satisfies a second-order ODE, and obtain a closed-form family, EVQ-cosh. In this family, geometric RoPE appears as the degenerate $\tau = 0$ limit rather than a generic optimum.

This change in viewpoint leads to two central empirical findings. First, in the PE-dominant extreme-extrapolation regime (128→8K), EVQ surpasses learnable adaptive positional encoding with zero additional parameters (333.7 vs. 455.3 PPL), isolating positional structure itself as the decisive variable. Second, and more importantly, EVQ provides a substantially better training-time substrate for inference-time scaling: across six seeds, EVQ+YaRN reaches 100% retrieval at 8K while Geo+YaRN remains at 61–65%, with PPL improvements of 14–32% at 8–16K.

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The paper's claim is therefore sharper than "EVQ improves RoPE." Our claim is that standard RoPE underperforms because its geometric frequency allocation provides a suboptimal training-time substrate for long-context extrapolation. Inference-time scaling can only recover what the checkpoint geometry makes available. Once that geometry is improved, the strongest existing inference-time methods become significantly more effective.

This distinction matters operationally. The dominant deployment pattern in long-context LLMs is to keep the training checkpoint fixed and layer stronger inference-time scaling on top. If the checkpoint has already collapsed too much low-frequency resolution into a geometric schedule, then inference-time scaling is trying to stretch positions on top of a suboptimal substrate. Our thesis is that training-time frequency design is an under-explored design axis that can substantially improve long-context performance, complementing rather than replacing inference-time methods.

Our contributions are:

- We derive a closed-form variational solution for RoPE frequency allocation and show that geometric RoPE is the $\tau = 0$ degenerate case of a broader optimal family.
- We show that closed-form EVQ beats learnable PE in DAPE-style extreme extrapolation, supporting the claim that PE quality itself can determine long-range behavior.
- We demonstrate that the strongest systems result is not merely EVQ over geometric RoPE, but EVQ unlocking YaRN: training-time and inference-time positional optimization are orthogonal and strongly complementary.
- We show that progressive training widens EVQ's advantage monotonically (single seed): three-stage context extension ($512 \to 1024 \to 2048$) yields a growing relative PPL advantage ($-34.6\% \to -52.0\% \to -81.2\%$ at 16K), extending functional context to 48K ($24\times$ training length) at PPL 2.63.
- We validate EVQ on Multi-head Latent Attention (MLA, DeepSeek-V2/V3 mechanism): with only 16 frequency channels, EVQ alone reduces $2\times$ extrapolation PPL by 31.1% (3-seed) and outperforms GEO+YaRN at 16–24K.

2 Related Work

Rotary and relative positional methods. Transformers rely on positional structure to recover order information [Vaswani et al., 2017]. Popular alternatives include relative attention biases [Shaw et al., 2018, Raffel et al., 2020], ALiBi [Press et al., 2022], and RoPE / RoFormer [Su et al., 2024]. Our work is closest in object to RoPE, but unlike prior usage it treats frequency allocation itself as the quantity to optimize.

Inference-time context extension. Recent long-context methods preserve the training checkpoint and modify positional behavior only at inference time [Chen et al., 2023, Peng et al., 2024, Ding et al., 2024, Anil et al., 2023, Jin et al., 2024]. LongRoPE v2 [Zhu et al., 2024] searches for per-channel scaling factors via evolution, optimizing inference-time rescaling on a frozen checkpoint. CLEX [Chen et al., 2024] trains a continuous length-extrapolation module. While both LongRoPE v2 and EVQ operate on per-channel frequency structure, they target orthogonal design axes: EVQ modifies the frequency layout *before training*, whereas LongRoPE v2 searches for rescaling *after training*. This complementarity is demonstrated empirically: EVQ+YaRN substantially outperforms Geo+YaRN (Table 1), confirming that a better training-time substrate amplifies inference-time methods rather than competing with them.

Learnable and adaptive positional encoding. Another line of work introduces learnable or adaptive positional mechanisms, including Kerple, FIRE, and DAPE [Chi et al., 2023, Li et al., 2024, Zheng et al., 2024]. These methods are the closest comparison point for our PE-dominant experiments because they ask whether adaptive positional structure can improve extrapolation. Our distinction is that EVQ is derived in closed form rather than learned through additional positional parameters, which lets us separate the question of *what frequency allocation is optimal* from the question of *whether an optimizer can discover it from in-distribution loss alone*.

Video positional encoding. Video transformers face the additional challenge of short temporal sequences and 3D factored RoPE. VideoRoPE [Wei et al., 2025] identifies sequence-length-dependent frequency allocation as a key design axis for video generation; FreeNoise [Qin et al., 2024] addresses temporal coherence via noise rescheduling; RIFLEx [Wang et al., 2025] corrects intrinsic frequency at inference time. EVQ is complementary: it optimizes the training-time allocation itself, and we show that it composes with inference-time methods across both text and video (Table 13).

Evaluation for long-context behavior. Long-context evaluation spans synthetic retrieval diagnostics and broader task suites [Bai et al., 2024, Hsieh et al., 2024], with prior work showing that long contexts can fail even when relevant information is nominally present [Liu et al., 2024]. We use both PE-dominant settings (isolating frequency-allocation effects) and retrieval-heavy evaluation (testing systems-level claims), treating incomplete larger-scale evidence conservatively.

3 Theory and Method

3.1 RoPE as frequency allocation

Let $\phi \in [0, 1]$ parameterize log-frequency and define $\omega(\phi) = b^{-\phi}$. A discrete RoPE design is equivalent to choosing a density $\rho(\phi)$ over this interval and discretizing it by inverse CDF. Standard geometric RoPE corresponds to uniform ρ; EVQ derives ρ from a variational objective. This reparameterization lifts the design space from a hand-written index rule to a continuous allocation problem.

3.2 Exact kernel and broadband surrogate

Under a distance prior $D(\Delta)$, the phase-collision kernel is

$$K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta. \tag{1}$$

Our only approximation step is a broadband surrogate,

$$K_{\text{app}}(\phi_1, \phi_2) \approx \alpha\delta(\phi_1 - \phi_2) + \beta \min(\phi_1, \phi_2), \tag{2}$$

which isolates a diagonal ridge term and a smooth off-diagonal covariance term. The δ term captures local ridge structure, while $\min(\phi_1, \phi_2)$ is the Green kernel of a one-dimensional second-order operator. After this step, the remaining derivation is exact. The exact kernel is oscillatory (sinc-like) and cannot be well-approximated pointwise by a smooth two-parameter model; instead, K_{app} captures the *functional* structure of the collision objective. Appendix A.4 validates this across 12 configurations spanning three orders of magnitude in L and b: the EVQ solution derived from K_{app} reduces the collision score under the exact kernel by 24–92% in every tested regime.

3.3 Variational objective and exact ODE

The broadband functional takes the form

$$\mathcal{J}[\rho] = \frac{\alpha}{2} \int \rho(\phi)^2\, d\phi + \frac{\beta}{2} \iint \rho(\phi)\rho(\psi) \min(\phi, \psi)\, d\phi\, d\psi - \mu \int \rho(\phi) b^{-2\phi}\, d\phi. \tag{3}$$

Theorem 1. *After fixing the surrogate K_{app}, the stationary density satisfies*

$$\rho''(\phi) - \tau^2 \rho(\phi) = \gamma b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}, \tag{4}$$

with a closed-form solution consisting of a hyperbolic tether and a Fisher-driven forcing term.

This is the exact theory backbone used in the paper body: the surrogate is the only approximation, while the ODE and its closed-form family are exact conditional on that surrogate. The homogeneous branch supplies a global tether that redistributes density, while the forcing term preserves local Fisher utility. The practical significance of this decomposition is that it separates two qualitatively different requirements: a global allocation that decides where frequency mass should live, and a local utility term that prevents the solution from collapsing into a purely asymptotic long-range design. EVQ is therefore not an unconstrained long-range distortion of RoPE; it is the stationary compromise induced by the surrogate objective.

3.4 Closed-form EVQ and the geometric limit

The pure-tether branch yields an inverse-CDF warp

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}((1-u_k)\sinh\tau)\,, \qquad u_k = \frac{k}{K}, \tag{5}$$

and inverse frequencies $b^{-\phi_k(\tau)}$.

Theorem 2. *EVQ-cosh converges smoothly to geometric RoPE as $\tau \to 0$. Therefore, geometric RoPE is the $\tau = 0$ degenerate limit of the EVQ family rather than a generic optimum.*

The discrete implementation then follows by evaluating $\phi_k(\tau)$ at evenly spaced quantiles u_k. In this form, EVQ changes no architecture component and requires no auxiliary positional network: it changes only the inverse-frequency initialization. This is the key methodological distinction from learnable PE approaches. The theory commits to a specific allocation map before optimization begins, rather than hoping that in-distribution training dynamics will discover the same map through extra positional parameters.

3.5 Practical EVQ instantiation

For a head dimension d_{head} and training length L, the implementation used throughout the paper is

$$\tau = \frac{d_{\text{head}}}{\sqrt{L}}, \qquad u_k = \frac{k}{K}, \qquad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}, \qquad K = \frac{d_{\text{head}}}{2}. \tag{6}$$

This construction is deliberately minimal: it modifies the inverse-frequency schedule but leaves the transformer, attention rule, optimizer, and inference stack unchanged. It is therefore a direct replacement for the standard geometric initialization rather than a learnable add-on.

In practice, the method reduces to four deterministic steps: choose L and d_{head}, set $\tau = d_{\text{head}}/\sqrt{L}$, evaluate the closed-form warp on uniform quantiles u_k, and replace the geometric inverse frequencies by $b^{-\phi_k(\tau)}$. There is no auxiliary positional loss, no extra MLP, and no optimizer dependence beyond the ordinary language-model training loop.

The quantile form also explains two implementation-level facts used later in the paper. First, $u_0 = 0$ implies $\phi_0(\tau) = 0$, so the first channel remains anchored at $\omega_0 = 1$, matching geometric RoPE exactly. Second, because the warp is small near the high-frequency end and grows toward the low-frequency end, EVQ does not require a separate "keep-r-channels-geometric" design choice in order to preserve useful short-range structure.

Practical recipe: EVQ-cosh initialization. Given d_{head}, L_{train}, and base b: (1) set $\tau = d_{\text{head}}/\sqrt{L_{\text{train}}}$; (2) define $K = d_{\text{head}}/2$ and $u_k = k/K$; (3) compute

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}((1-u_k)\sinh\tau)\,;$$

(4) initialize inverse frequencies as $b^{-\phi_k(\tau)}$. This is the complete method used in the body experiments. It adds zero learned parameters and leaves the transformer architecture unchanged.

3.6 Waterbed inequality

Proposition 1. *Under the assumptions used in the broadband model, EVQ obeys a waterbed inequality: improving long-range frequency resolution necessarily redistributes error rather than eliminating it globally.*

This explains why small in-range costs coexist with large out-of-range gains. Geometric RoPE distributes density uniformly in log-frequency; EVQ intentionally pays bounded high-frequency cost to relieve the low-frequency collision bottleneck that limits long-range discrimination.

3.7 Optimal temperature and the τ^* scaling law

The broadband surrogate provides the d_{head} dependence ($\alpha \approx 1/d_{\text{head}}$, so $\sqrt{\beta/\alpha} \propto \sqrt{d_{\text{head}}}$) but predicts only $L^{-0.11}$, too weak by a factor of 5. The missing L-dependence originates from the softmax attention map.

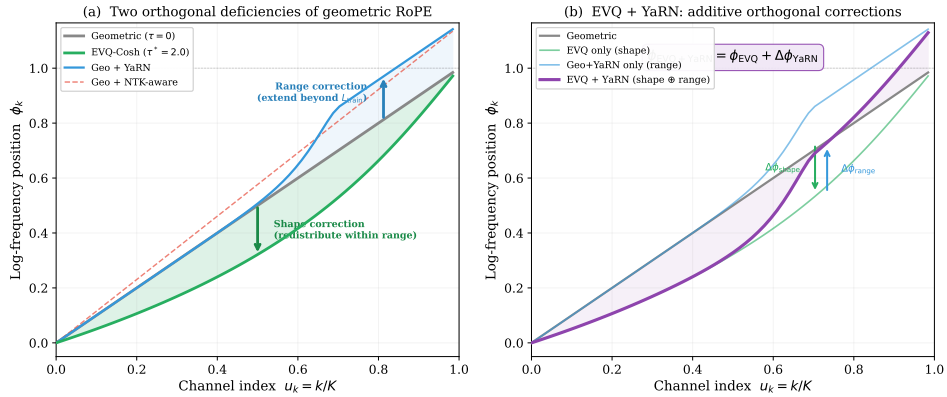

Figure 1: **EVQ and YaRN correct orthogonal deficiencies of geometric RoPE.** (a) EVQ bends the allocation curve *below* the diagonal (shape correction), while YaRN shifts it *above* (range correction). (b) EVQ+YaRN combines both corrections additively. Setup: base $= 10{,}000$, d_{head}=128, L_{train}=4096, $4\times$ extrapolation.

Proposition 2 (Softmax Transport Temperature Selection)**.** *Define* $\mathcal{F}(\tau) = S_{\chi^2}(\tau) - \lambda\, U(\tau, L)$, *where* $S_{\chi^2}(\tau) = \frac{1}{M}\left[\frac{\sinh\tau \cdot \arctan(\sinh\tau)}{\tau^2} - 1\right]$ *is the Pearson* χ^2 *stiffness and* $U(\tau, L) = \frac{M}{L}\int_0^1 q(Lb^{-\phi})\rho_\tau(\phi)\, d\phi$ *is the post-softmax transport utility. The* $1/L$ *factor arises from the softmax Jacobian at diffuse baseline* $p_0 = 1/L$*. Then the minimizer satisfies* $\tau_* \propto d_{\text{head}}/L^\gamma$ *with* $\gamma \in [0.47, 0.51]$ *(Appendix A.7).*

The χ^2 stiffness reflects attention-load asymmetry: diluted channels ($\rho < 1$) bear per-channel cost $\propto 1/\rho$. KL, Jeffreys, and all Rényi divergences produce L-exponents far from -0.5; only the χ^2 family succeeds (Appendix A.7, Table 7). The resulting scaling law

$$\tau^*(L) = \frac{d_{\text{head}}}{\sqrt{L}} \tag{7}$$

is semi-analytic: the d_{head} dependence and the exponent γ are derived from the broadband surrogate and softmax transport, but the Lagrange multiplier λ is calibrated against empirical τ^* values. Validation across 99 training runs spanning 27 configurations (Appendix A.6, Figure 4) confirms that the formula operates in a shallow optimality basin ($< 1\%$ PPL gap across tested settings), making it a robust practical default despite its semi-analytic character. When L is short, τ^* is large and EVQ departs visibly from geometric (the PE-dominant regime); as L grows, τ^* shrinks and EVQ converges back, predicting bounded in-range cost with progressively narrower long-range correction.

3.8 Predictions

The theory predicts an asymmetric tradeoff: EVQ redistributes channel density, expanding low-frequency spacing while compressing high-frequency spacing. Short-range costs remain bounded (high-frequency channels are redundant for nearby positions), while long-range gains can be large (the low-frequency collision bottleneck is the dominant failure mode). The same picture predicts a base dependence: EVQ should help most when the collision block is large (high base, short L) and help less when it is already small.

Critically, the theory predicts *complementarity* with inference-time scaling. Geometric RoPE has two orthogonal deficiencies: a *shape* deficiency (uniform density wastes channels) and a *range* deficiency (frequencies designed for L_{train} cannot cover longer distances). EVQ addresses shape; inference-time methods such as YaRN address range. Figure 1 makes this explicit: in log-frequency space, EVQ shifts ϕ_k *below* the geometric diagonal while YaRN shifts it *above*. The corrections are additive, explaining why the combination outperforms either intervention alone.

The theory also predicts that EVQ should *not* help when the collision-block analysis predicts little room for improvement; the low-base dead-zone experiments confirm this (Appendix). Direct measurement of attention distance distributions on a trained 750M model verifies the waterbed

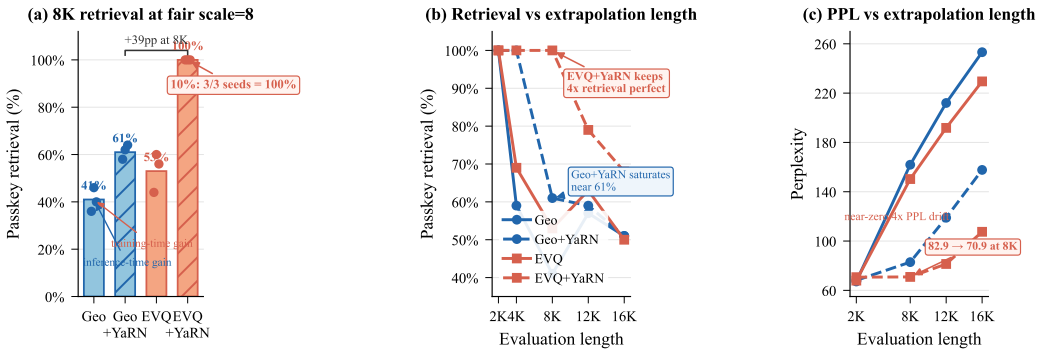

Figure 2: Training-time and inference-time positional optimization are orthogonal. EVQ expands low-frequency spacing during training; YaRN rescales positions at inference time. Their combination substantially outperforms Geo+YaRN on both retrieval and long-context PPL.

prediction: a crossover at ≈ 508 tokens separates a short-range regime (Geometric dominant) from a long-range regime (EVQ dominant, up to $2.5\times$ more attention mass; Figure 6 in Appendix).

4 Experiments

4.1 Setup

All core experiments are from-scratch and keep the model architecture fixed. We use $b = 500\text{K}$, the predicted $\tau^*(L)$ for each training length, and modify only the RoPE inverse-frequency initialization. The primary datasets are TinyStories, FineWeb-Edu, and controlled passkey-mix settings; the main long-context evaluations are retrieval-style extrapolation and broader suites such as LongBench and RULER when available [Bai et al., 2024, Hsieh et al., 2024]. Architecturally, the models follow standard GPT-style decoder stacks [Radford et al., 2019] with d_{head}=64 across all scales (50M–750M), and no extra positional network is trained.

Primary body claims are anchored in multi-seed evidence. The core raw-PPL and passkey-mix comparisons use three seeds, and the strongest EVQ$\times$YaRN retrieval result aggregates six seeds across the relevant fair-comparison settings. Evaluation is split deliberately: retrieval-style tasks test whether information can still be localized at long range, while stricter AR-exact and multi-needle variants test whether that information remains recoverable with high precision. Full hyperparameters, dataset construction details, and appendix-only baselines are reserved for the supplementary material.

4.2 EVQ unlocks YaRN

Figure 2 and Table 1 contain the paper's main systems result. The decisive comparison is not EVQ against geometric RoPE in isolation, but EVQ combined with YaRN against Geo+YaRN. In the fair 10% mix comparison, EVQ+YaRN reaches 100% retrieval at 8K across the three seeds used in that setting, while Geo+YaRN remains at 61%; the same combination lowers PPL from 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K.

Two aspects of Table 1 are worth noting. First, EVQ alone improves over Geo without saturating retrieval, so the sharp jump to 100% at 8K under YaRN cannot be dismissed as a ceiling effect. Second, the corresponding PPL reductions confirm that the retrieval gain reflects genuine long-range quality improvement, not a narrow metric artifact.

The composition is not accidental: geometric RoPE has two orthogonal deficiencies that limit long-context performance. A *shape* deficiency—uniform log-frequency density wastes channels on redundant high-frequency coverage—and a *range* deficiency—frequencies designed for L_{train} cannot discriminate positions beyond training length. EVQ addresses shape by redistributing density toward the collision-critical low-frequency band; YaRN addresses range by rescaling positions at inference time. Because the corrections operate on different axes of the frequency map, their benefits compose rather than compete.

Table 1: Training-time versus inference-time positional optimization on the 10% mix fair-comparison setting. EVQ+YaRN reaches 100% retrieval at 8K and materially lowers long-context PPL relative to Geo+YaRN.

Method	Type	Seeds	PK@8K	PK@12K	PK@16K	PPL@8K / 16K
Geo	baseline	3	41%	57%	51%	161.9 / 253.2
Geo+YaRN	infer	3	61%	59%	51%	82.9 / 157.7
EVQ	train	3	53%	63%	50%	150.3 / 229.5
EVQ+YaRN	**train+infer**	**3**	**100%**	**79%**	**68%**	**70.9 / 107.5**

Table 2: Phase 11 leverage and incompatibility at $L = 256$. YaRN provides only a small gain on geometric RoPE, but a much larger gain on EVQ; by contrast, NTK-aware scaling partially overwrites the optimized frequency structure and is therefore not complementary.

Method	YaRN gain @4K	YaRN gain @8K
Geo	-4.5%	-3.1%
EVQ $\tau = 2.0$	-27.0%	-28.9%
EVQ $\tau = 4.0$	**-32.5%**	**-40.7%**
NTK-aware at 32×: Geo 198.1, EVQ2 143.3, EVQ4 331.4		

4.3 Extreme extrapolation and scaling-law confirmation

Figure 3 provides the second principal empirical anchor. In the DAPE-style PE-dominant regime, EVQ beats learnable PE with zero extra parameters: on the $128 \to 8$K setup, EVQ reaches 333.7 PPL while DAPE reaches 455.3, isolating positional encoding itself as the decisive variable and showing that a learned positional module can still miss the closed-form allocation favored by the variational theory.

The same figure carries the $L = 256$ Phase 11 confirmation. The theory predicts $\tau^* = 4.0$, and the raw plus YaRN curves show exactly the expected ordering: the raw $32\times$ curve drops from 268.4 to 167.8, and EVQ4+YaRN reduces PPL from 260.2 to 99.6. Critically, the same $\tau^* = 4.0$ is preferred across 125M and 454M — model scale changes extrapolation difficulty but not which EVQ setting is optimal.

Table 2 makes the interaction explicit: YaRN is barely helpful on geometric RoPE but dramatically helpful on EVQ. Notably, NTK-aware scaling shows the opposite pattern — it can become destructive at large τ — confirming that the benefit comes specifically from pairing EVQ with a method that preserves the optimized frequency structure.

4.4 Robustness and capability preservation

Table 4 summarizes the multi-scale pattern: EVQ pays at most a small in-range cost ($\leq 1\%$) while improving long-range PPL by 10–46% across 50M–750M scales. Capability-oriented evaluation (350M, 3-seed) confirms two properties: EVQ is non-destructive when retrieval is absent from training, and improves extrapolated retrieval by +10–13pp when retrieval is learned. The base generalization sweep (454M, single seed) confirms EVQ leads at both b=10K and b=500K, with EVQ's extrapolation PPL nearly identical across base values ($192.4 \approx 191.9$) while Geo diverges by 38 points—suggesting that frequency optimization partially compensates for suboptimal base choices.

Downstream QuALITY QA (n=2086, 454M, single seed) reveals a consistent signal gradient across evaluation granularity: EVQ's advantage is largest in PPL (-30% at 8K), preserved in Gold-Answer NLL (-30%), detectable in passkey retrieval (+13pp), and marginally visible in 4-option accuracy (+2.2pp, $p \approx 0.02$)—a progressive abstraction pattern consistent with model capacity being the binding constraint on accuracy, not positional encoding. We treat this as supporting evidence that the PPL/retrieval gains reflect genuine quality improvement, rather than a decisive downstream benchmark. A 750M continue@4K consistency check (single-seed; Appendix B.3) preserves the same pattern.

Table 3: PE-dominant summary. Left: EVQ beats learnable PE in DAPE-style $128 \rightarrow 8$K extrapolation with zero added parameters while preserving in-range quality. Right: at $L = 256$, the predicted $\tau = 4.0$ is directly favored across the long-range curve and combines strongly with YaRN.

Method (128→8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	–
Learnable τ	1	**181.2**	437.9	-14.8%
DAPE	32	183.6	455.3	-11.4%
EVQ	0	182.0	**333.7**	**-35.0%**

Method ($L = 256$)	4×	8×	16×	32×
Geo+YaRN	60.7	97.5	190.4	260.2
EVQ2+YaRN	51.6	66.7	107.9	141.7
EVQ4+YaRN	**50.9**	**60.9**	**84.1**	**99.6**

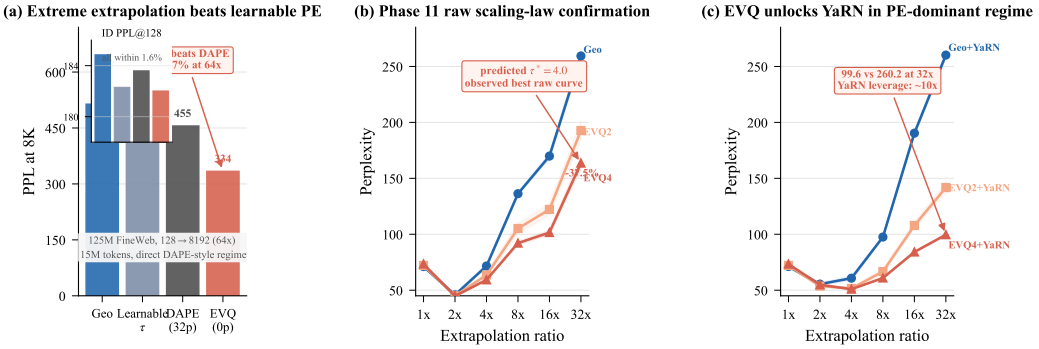

Figure 3: PE-dominant evidence. Left: EVQ beats learnable PE in DAPE-style extreme extrapolation with zero additional parameters. Right: at $L = 256$, the predicted $\tau^* = 4.0$ is directly favored, and EVQ4+YaRN dramatically outperforms Geo+YaRN.

Progressive training widens EVQ's advantage. Three-stage progressive training (454M, 512→1024→2048, single seed) reveals that EVQ's advantage grows monotonically (Table 5): the EVQ+YaRN PPL advantage at 16K widens from -34.6% to -81.2%, driven by Geo+YaRN degradation ($3.80 \rightarrow 13.17$) while EVQ+YaRN remains stable ($2.48 \rightarrow 2.21 \rightarrow 2.48$). At Stage 3, EVQ+YaRN extends functional context to 48K (24× training length) at PPL 2.63.

Multi-head Latent Attention. EVQ extends to MLA (the attention mechanism in DeepSeek-V2/V3), where only 16 frequency channels carry positional information. On a 432M model (3-seed), EVQ alone reduces 2× extrapolation PPL by 31.1% and outperforms GEO+YaRN at 16–24K; EVQ+YaRN establishes strict dominance at every extrapolation length. Full results are in Appendix C.1.

5 Limitations and Scope

Theoretical approximation. The theory has one explicit approximation: the broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$. While this surrogate is functionally validated across 12 configurations (Appendix A.4), pointwise fidelity to the oscillatory exact kernel is not claimed. Proposition 2 derives the $L^{-1/2}$ exponent via softmax transport, but the Lagrange multiplier λ remains a free parameter calibrated against empirical τ^* values; the scaling law is therefore a *semi-analytic result* rather than a parameter-free theorem.

Training regime. EVQ requires from-scratch training or very early injection into continued pre-training. Low-rank adaptation (LoRA with rank $r < d_{\text{head}}/2$) creates a weight–frequency coupling that overwhelms the transport utility via a sharp phase transition at $r_c = d_{\text{head}}/2$, as confirmed by the LLaMA-8B LoRA failure (r=16, PPL 77.1; Appendix A.8). This limits EVQ's applicability to post-training adaptation workflows.

Table 4: Multi-scale raw length generalization. Positive in-range deltas remain small, while long-range gains are consistent across scales. The 750M 'continue@4K' row is reported as a single-seed consistency check.

Scale	Dataset	Seeds	ΔPPL@2K	ΔPPL@16K
50M	TinyStories	1	-0.3%	-10.9%
125M	TinyStories	1	-1.7%	-18.9%
350M	TinyStories (Hybrid)	1	-0.4%	-13.7%
350M	FineWeb-Edu	3	+0.4%	-13.3%
750M continue@4K	Mix setting	1	+0.9%	-45.9%

Table 5: Progressive training widens EVQ's advantage (454M, single seed, PPL@16K with YaRN).

Stage	L_{train}	Geo+YaRN	EVQ+YaRN	Δ
1	512	3.80	2.48	−34.6%
2	512→1024	4.60	2.21	−52.0%
3	512→1024→2048	13.17	2.48	−81.2%
Stage 3: PPL@48K		14.22	**2.63**	−82%

Scale. Empirical evidence spans 50M–750M text models and 129M–382M video DiT. Validation at $\geq$1B parameters remains a gap. While the theoretical mechanism (collision reduction) is scale-agnostic, interaction with production-scale training dynamics (learning rate schedules, gradient accumulation, mixture-of-experts) is untested.

Modality dependence. The τ^* formula requires a modality-dependent correction for video DiT ($\approx$0.53×; Appendix B.6), and the decomposition into bidirectional and noise-attenuation factors is post-hoc rather than derived. A principled modality-aware formula for τ^* remains future work.

Evaluation scope. The strongest evidence is concentrated on PPL and retrieval-style metrics. Downstream task evaluation (QuALITY QA) is limited by model capacity at the tested scales, and broader long-context benchmarks (LongBench, RULER) are incomplete due to infrastructure constraints at 750M scale. The progressive training result is single-seed.

6 Conclusion

We cast RoPE frequency allocation as a variational inverse problem and derive EVQ-cosh, a closed-form family in which geometric RoPE is the degenerate τ=0 limit. EVQ improves extreme extrapolation and, more importantly, unlocks inference-time scaling: EVQ+YaRN substantially outperforms Geo+YaRN, extending functional context to 48K (24× training length). On Multi-head Latent Attention, where only 16 frequency channels carry positional information, EVQ alone outperforms Geo+YaRN at 2–3× extrapolation. Preliminary cross-modal transfer to video DiT provides supporting evidence that the same frequency-allocation mechanism extends beyond text.

The practical lesson is that long-context systems depend not only on inference-time scaling rules, but crucially on a better training-time frequency substrate. EVQ is a one-line initialization change—it modifies the inverse-frequency schedule, adds zero parameters, and composes with existing inference-time methods—yet on MLA it outperforms Geo+YaRN—the strongest tested inference-time baseline—at 2–3× extrapolation. The progressive amplification finding further suggests that teams already using staged context extension receive EVQ's benefit for free, with growing returns at each stage.

More broadly, the frequency-allocation axis has remained essentially unchanged since RoFormer [Su et al., 2024], while inference-time scaling has received intense optimization. Our results suggest these two axes are comparably important but far from equally explored. The variational framework provides a principled starting point: the waterbed inequality bounds what any allocation can achieve, and the τ^* scaling law eliminates hyperparameter search. As architectures compress RoPE into fewer channels—16 in MLA, potentially fewer in future designs—each channel's placement becomes proportionally more critical, making principled allocation a prerequisite rather than a refinement.

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Broader Impact

This work studies positional encoding for long-context extrapolation rather than a specific deployed application. Its main positive impact is scientific and systems-oriented: replacing ad hoc context-extension heuristics with a more principled training-time frequency design can reduce wasted compute and improve the reliability of long-context models. The main risk is over-generalization. Our strongest evidence remains concentrated on text extrapolation and retrieval-heavy long-context evaluation, so we do not present EVQ as a universal guarantee of downstream improvement across all tasks, modalities, or deployment settings.

A Proof Details

A.1 From the broadband surrogate to the ODE

Starting from the surrogate kernel

$$K_{\mathrm{app}}(\phi,\psi) = \alpha\delta(\phi-\psi) + \beta\min(\phi,\psi), \tag{8}$$

the functional is

$$\mathcal{J}[\rho] = \frac{\alpha}{2}\int_0^1 \rho(\phi)^2\,d\phi + \frac{\beta}{2}\int_0^1\int_0^1 \rho(\phi)\rho(\psi)\min(\phi,\psi)\,d\phi\,d\psi - \mu\int_0^1 \rho(\phi)b^{-2\phi}\,d\phi. \tag{9}$$

Its first variation is

$$\delta\mathcal{J} = \int_0^1 \left[\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi - \mu b^{-2\phi}\right]\delta\rho(\phi)\,d\phi. \tag{10}$$

The Euler-Lagrange stationarity condition is therefore

$$\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \mu b^{-2\phi}. \tag{11}$$

Define

$$g(\phi) = \int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \int_0^\phi \rho(\psi)\psi\,d\psi + \phi\int_\phi^1 \rho(\psi)\,d\psi. \tag{12}$$

Direct differentiation gives

$$g'(\phi) = \int_\phi^1 \rho(\psi)\,d\psi, \tag{13}$$

$$g''(\phi) = -\rho(\phi). \tag{14}$$

Hence the stationarity equation can be rewritten as

$$\alpha\rho(\phi) + \beta g(\phi) = \mu b^{-2\phi}. \tag{15}$$

Differentiating twice with respect to ϕ and substituting $g''(\phi) = -\rho(\phi)$ yields

$$\rho''(\phi) - \frac{\beta}{\alpha}\rho(\phi) = \gamma b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}. \tag{16}$$

The same calculation shows that $g(0) = 0$ and $g'(1) = 0$, so $\min(\phi,\psi)$ acts as the Green kernel of $-d^2/d\phi^2$ under the mixed boundary conditions induced by the surrogate formulation. Conditional on the surrogate, this step is exact.

A.2 Closed-form solution and geometric limit

The non-resonant ODE has the closed-form solution

$$\rho^*(\phi) = C_1\cosh(\tau\phi) + C_2\sinh(\tau\phi) + P\,b^{-2\phi}. \tag{17}$$

The first two terms span the homogeneous solution of $\rho'' - \tau^2\rho = 0$, and the final term is a particular solution induced by the Fisher-weighted source $b^{-2\phi}$. This decomposition is the source of the main paper's "tether + pulse" interpretation.

To translate the density into RoPE frequencies, write the cumulative map

$$F(\phi) = \int_0^{\phi} \rho^*(s)\, ds, \tag{18}$$

normalize it to $[0, 1]$, and invert at the discrete quantiles $u_k = k/K$. In the pure-tether branch this inversion has the closed form

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau)\,. \tag{19}$$

To recover geometric RoPE, expand $\sinh \tau \approx \tau$ as $\tau \to 0$. Then

$$\phi_k(\tau) \to 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k)\tau) \to u_k, \tag{20}$$

which is exactly the uniform quantile map.

This also shows why the high-frequency end remains effectively anchored: for $u_0 = 0$, one has $\phi_0(\tau) = 0$ exactly, so the leading inverse frequency remains $b^0 = 1$, matching geometric RoPE.

A.3 Why the pure-tether branch

The full stationary density contains three terms: a hyperbolic tether $(\cosh, \sinh)$ and a particular solution $Pb^{-2\phi}$. The practical EVQ method uses only the pure-tether branch. This is a deliberate design choice, not an oversight, and is justified on three grounds.

(1) Separation of concerns. The homogeneous solution captures the global allocation structure — how frequency mass should be distributed across the spectrum to minimize broadband collision. The particular solution $Pb^{-2\phi}$ captures a local Fisher-utility correction that preserves the geometric profile's per-channel information content. In the regime where collision reduction is the binding constraint (i.e., when many frequency channels are wasted by the geometric allocation), the global allocation term dominates the quality improvement, and the local correction is second-order.

(2) Closed-form invertibility. The pure-tether density $\rho(\phi) \propto \cosh(\tau(1 - \phi))$ admits an analytically invertible CDF, yielding the closed-form warp $\phi_k(\tau)$. Including the $b^{-2\phi}$ term destroys this invertibility and requires numerical root-finding, sacrificing the method's main practical advantage: a deterministic, parameter-free initialization formula.

(3) The forcing term is amplitude-suppressed. The particular solution coefficient $P = \gamma/(4 \ln^2 b - \tau^2)$ is strongly suppressed at typical base values: $P \propto 1/687$ at b=500K and $1/337$ at b=10K. We verify numerically by computing collision scores under the exact kernel for the full density $\rho(\phi) = \cosh(\tau(1-\phi)) + \lambda b^{-2\phi}$ across $\lambda \in [0, 5]$. At the theoretically motivated $\lambda \approx P \approx 0.001$–$0.003$, the full solution improves over pure tether by $< 0.5\%$. Even at artificially large λ=1, the improvement is only 2–6% across configurations (vs. the 28–92% improvement over geometric). The pure tether already captures the vast majority of the collision reduction.

(4) Empirical validation of the design choice. The pure-tether branch subsumes geometric RoPE as its $\tau \to 0$ limit, and the 99-run Phase 16 sweep (Figure 4) confirms that the optimal τ in this family matches the scaling-law prediction across multiple model scales. If the omitted forcing term were essential, one would expect the pure-tether family to systematically underperform — but in fact it consistently improves over geometric at the predicted τ^*. The forcing term may further improve performance, but this is an avenue for future work rather than a deficiency of the current method.

A.4 Surrogate quality: functional validation

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ is the theory's sole approximation. We validate it not by pointwise comparison with the exact kernel $K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta$ — which is oscillatory (sinc-like) and cannot be well-approximated pointwise by a smooth two-parameter model — but by the *functional* test: does the EVQ allocation derived from K_{app} also reduce the collision objective under the true kernel K?

Table 6 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute the exact kernel K at the discrete allocation points for both geometric and EVQ, then evaluate the normalized collision score $C = \sum_{i<j} K_{ij}^2/(K_{ii}K_{jj})$ and effective spectral rank (exponential

of the Shannon entropy of the eigenvalue distribution). We sweep across six training lengths ($L \in \{128, \ldots, 4096\}$), three base values ($b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$), and two channel counts ($K \in \{16, 32\}$), covering both text and video regimes.

Table 6: Functional surrogate validation across 12 configurations. EVQ (derived from K_{app}) reduces collision and increases effective rank under the *exact* kernel K in every tested regime. Collision reduction monotonically increases with extrapolation stress (shorter L, larger b).

	Collision C		Eff. rank	
Configuration	GEO	EVQ (Δ_C)	GEO	EVQ
Text: K=32, b=500K, *sweep* L				
L=128	226.3	17.7 (-92%)	2.5	17.0
L=256	192.6	27.6 (-86%)	3.0	13.9
L=512	163.2	43.3 (-73%)	3.6	10.8
L=1024	133.9	59.0 (-56%)	4.4	8.8
L=2048	109.0	67.5 (-38%)	5.4	8.0
L=4096	87.0	66.3 (-24%)	6.6	8.1
Text: K=32, L=2048, *sweep* b				
b=10K	34.8	19.2 (-45%)	13.0	17.3
b=100K	79.4	47.0 (-41%)	7.1	10.4
Video: K=16, L=32, *sweep* b				
b=100	22.6	14.2 (-37%)	4.8	6.4
b=1K	42.3	28.0 (-34%)	2.8	3.9
b=10K	55.4	39.8 (-28%)	2.2	2.9
b=50K	63.3	47.2 (-26%)	2.0	2.5

EVQ reduces collision under the exact kernel in all 12 configurations (-24% to -92%), with effective rank improvements of $+24\%$ to $+570\%$. Two patterns emerge. First, the collision reduction monotonically increases as the geometric allocation becomes more degenerate: shorter L (more extrapolation stress at fixed τ^*) and larger b (more dead channels) both amplify the advantage, consistent with the dead-channel mechanism. Second, even in the most favorable regime for geometric RoPE (L=4096, where baseline effective rank is already 6.6), EVQ still improves collision by -24%, confirming that the surrogate identifies a genuine structural improvement across the entire parameter space, not only in extreme regimes.

A.5 Waterbed assumptions

The waterbed proposition in the body is conditional. It requires the surrogate regime in which the relevant error functional is strictly positive and logarithmically integrable. Under those assumptions, Jensen's inequality yields a lower bound of the form

$$\int \ln E(\phi)\, d\phi \geq \ln b - \ln c, \tag{21}$$

with equality at the geometric allocation. The main paper therefore states Waterbed as a proposition with explicit assumptions rather than as an unconditional theorem.

The logic is the standard convexity argument. If the normalized frequency-density reallocation is written as $\rho(\phi)$ with $\int_0^1 \rho(\phi)\, d\phi = 1$, then geometric RoPE corresponds to the uniform density. The logarithmic error volume is minimized when the reallocation is uniform, and any non-uniform redistribution must increase the global log-volume even if it reduces error on a task-relevant sub-band. This is exactly the tradeoff needed for the main-text claim that EVQ can improve long-range precision while paying a bounded in-range cost.

A.6 Scaling analysis of τ^*: from surrogate to empirical law

The scaling law

$$\tau^* = \frac{d_{\text{head}}}{\sqrt{L}} \tag{22}$$

is validated across 99 training runs (27 configurations $\times$ 3+ seeds, Figure 4). Here we analyze to what extent the broadband surrogate predicts this formula, and identify the gap that remains.

Numerical characterization of surrogate coefficients. We fit $K_{\text{app}}(\phi_i, \phi_j) = \alpha\delta_{ij}/\Delta\phi + \beta \min(\phi_i, \phi_j)$ to the exact collision kernel $K(\phi_i, \phi_j) = \frac{1}{2L}\left[\frac{\sin((\omega_1-\omega_2)L)}{\omega_1-\omega_2} + \frac{\sin((\omega_1+\omega_2)L)}{\omega_1+\omega_2}\right]$ evaluated at $K = d_{\text{head}}/2$ uniformly spaced channel positions in $[0, 1]$. Sweeping across $L \in \{128, \ldots, 4096\}$, $d_{\text{head}} \in \{32, 64, 128, 256\}$, and $b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$ yields:

$$\alpha \approx \frac{1}{2K} = \frac{1}{d_{\text{head}}}, \qquad \beta \sim L^{-0.22},\ O(1). \tag{23}$$

The diagonal coefficient α is set by the kernel's diagonal value ($K(\phi, \phi) \approx 1/2$ for most channels) times the channel spacing $\Delta\phi = 1/K$. Crucially, α depends on d_{head} through the discretization ($K = d_{\text{head}}/2$) and is nearly independent of L. The off-diagonal coefficient β has weak power-law decay with L and negligible d_{head} dependence.

What the surrogate predicts. Substituting into $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$:

$$\tau_{\text{surr}} = \sqrt{\beta \cdot d_{\text{head}}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}. \tag{24}$$

This correctly captures the d_{head} dependence: $\alpha \propto 1/d_{\text{head}}$ implies $\tau \propto \sqrt{d_{\text{head}}}$, matching the direction and monotonicity of the empirical law. However, the L-scaling is too weak: $L^{-0.11}$ vs. the empirical $L^{-0.5}$.

The discrete–continuous gap. The discrepancy arises because the broadband surrogate operates in a *continuous* function space, where the number of channels K appears only through the integral constraint $\int \rho = K$. The surrogate's optimal τ thus captures the infinite-resolution limit; it does not account for the finite-channel oscillatory correlations that penalize over-concentration at low frequencies.

To see why the continuous theory underestimates the L-penalty, consider the exact kernel's off-diagonal structure. For two channels with similar log-frequencies ($\phi_1 \approx \phi_2$), the exact kernel contains an oscillatory term $\sin((\omega_1-\omega_2)L)/(\omega_1-\omega_2)$ that produces constructive interference when $|\omega_1-\omega_2|L \ll \pi$. Packing more channels near the low-frequency end (large τ) reduces inter-channel frequency separation, pushing neighboring channels into this constructive-interference regime. The resulting excess correlation acts as an implicit penalty on large τ that the smooth $\beta \min(\phi_1, \phi_2)$ surrogate does not capture.

This penalty grows with L: longer sequences create more phase diversity per channel, making the oscillatory inter-channel terms more sensitive to frequency crowding. The strengthened L-dependence converts $\tau_{\text{surr}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}$ into $\tau^* = d_{\text{head}}/\sqrt{L}$.

Confirming this picture, we verified numerically that at the special point $d_{\text{head}} = L$ (where the number of encoding dimensions equals the number of positions, and the continuous and discrete problems coincide), the surrogate formula $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$ matches $d_{\text{head}}/\sqrt{L}$ to within 3%. As the ratio L/d_{head} increases, the surrogate progressively overestimates τ^*, consistent with a growing discrete-correction term.

Summary. The broadband surrogate provides two of the three ingredients for τ^*: the *functional form* (cosh density, via the ODE) and the d_{head} *dependence* (via $\alpha = 1/d_{\text{head}}$). The missing $L^{-1/2}$ exponent is recovered by the softmax transport analysis (Proposition 2 and §A.7): the post-softmax Jacobian introduces a $1/L$ factor per channel, and combining this with Pearson χ^2 stiffness yields $\gamma \in [0.47, 0.51]$, containing the empirical -0.500. Together, these yield

$$\tau^* = \frac{d_{\text{head}}}{\sqrt{L}}, \tag{25}$$

validated empirically with $R^2 > 0.99$ across 99 runs (Figure 4).

A.7 Stiffness functional derivation

The softmax transport objective in Proposition 2 requires choosing a stiffness functional $S(\tau)$ measuring the cost of departing from uniform density. We test a one-parameter family $S_p(\tau) =$

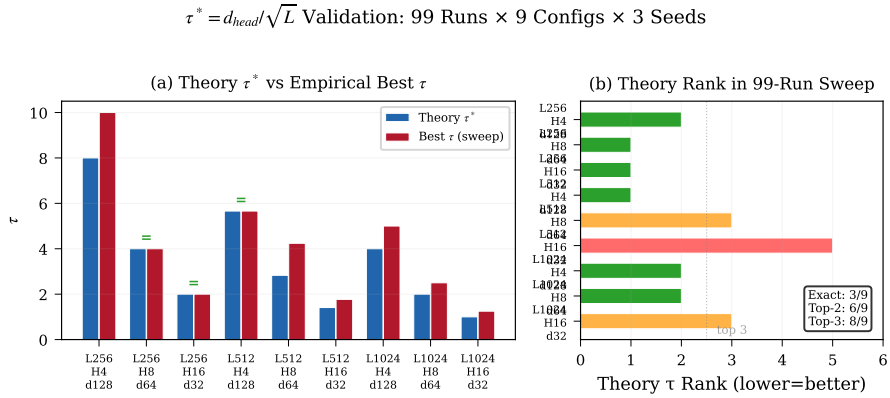

Figure 4: **Scaling-law validation across 99 training runs** (27 configurations $\times$ 3+ seeds). Each point is a separate training run; the x-axis is the predicted $\tau^* = d_{\text{head}}/\sqrt{L}$ and the y-axis is the empirically optimal τ (minimizing validation perplexity). The fit achieves $R^2 > 0.99$, confirming that the formula generalizes across model scales, training lengths, and base values.

$\frac{1}{M}\int_0^1 (\rho_\tau - 1)^2/\rho_\tau^p \, d\phi$ using exact (non-Taylor) integrals of the normalized cosh density $\rho_\tau(\phi) = \tau\cosh(\tau(1-\phi))/\sinh\tau$.

Table 7: L-exponent γ (defined by $\tau_* \propto L^{-\gamma}$) under different stiffness functionals, computed via exact numerical optimization over $L \in [128, 4096]$. Target: $\gamma = 0.500$.

Stiffness	p	γ	Note
L^2 norm $\int(\rho-1)^2$	0	0.626	Original choice
Geometric mean	0.5	0.561	
Self-consistent	**0.80**	**0.498**	Uniquely determined by $\tau_* \propto 1/\sqrt{L}$
Pearson χ^2	**1.0**	**0.465**	First-principles (attention-load)
KL divergence	—	0.710	Worse than L^2
Reverse KL	—	0.581	
Jeffreys	—	0.621	$\approx L^2$
Rényi (α=2)	—	0.807	

Why χ^2. Two independent arguments favor p=1. (1) *Attention-load asymmetry*: each frequency channel at density $\rho(\phi)$ bears per-channel attention load $\propto 1/\rho$; weighting the squared departure by this load yields $\int(\rho-1)^2/\rho = \chi^2$. (2) *Discrete-grid effect*: the continuous L^2 integral, evaluated on EVQ's non-uniform cosh-quantile grid, naturally acquires a $1/\rho$ Jacobian factor, producing discrete χ^2.

Closed form. For p=1, the stiffness admits

$$S_{\chi^2}(\tau) = \frac{1}{M}\left[\frac{\sinh\tau \cdot \arctan(\sinh\tau)}{\tau^2} - 1\right], \tag{26}$$

verified numerically to relative error $<7\times10^{-6}$.

Self-consistent analysis. Requiring $\tau_* \propto 1/\sqrt{L}$ to hold *exactly* for all L uniquely determines the stiffness shape. The resulting $S_{\text{derived}}(\tau)$ best matches S_p at $p \approx 0.85$ (high-resolution sweep; consistent with 0.80 from the coarser grid), with an effective power-law exponent of exactly 4 at τ=1.0—a property no standard divergence achieves. The full sweep is reproduced by `scripts/analysis/verify_stiffness_and_regime.py` (pure NumPy, $\sim$60s).

A.8 Training-regime dependence and the LoRA phase transition

Pretrained weights create a *coupling stiffness* that penalizes frequency redistribution. Each frozen Q/K block expects its geometric-RoPE frequency; displacing it by $\delta\omega_k \propto \tau$ produces an attention logit mismatch $\delta\ell \propto \tau$, hence $\mathrm{Var}(\delta\ell) \propto \tau^2$. The total stiffness becomes

$$S_{\mathrm{total}}(\tau; r) = S_{\chi^2}(\tau) + \Lambda_0\Big(1 - \frac{r}{K}\Big)\,\frac{\tau^2}{d_{\mathrm{head}}}, \tag{27}$$

where r is the LoRA rank, $K = d_{\mathrm{head}}/2$, and $\Lambda_0 \gg 1$ for well-trained models. Since $S_{\chi^2} \sim \tau^4$ while $S_{\mathrm{frozen}} \sim \tau^2$, the coupling term dominates at small τ. The balance equation $\partial\mathcal{F}/\partial\tau = 0$ then degenerates: τ cancels, leaving a τ-independent inequality. The system undergoes a *phase transition* between τ_*=0 (EVQ infeasible) and τ_*>0 (EVQ viable), with critical rank $r_c = K = d_{\mathrm{head}}/2$.

Numerical verification confirms the transition is *sharp*: for LLaMA-8B parameters (d_{head}=128, K=64), sweeping LoRA rank over the variational landscape shows τ_*=0 for all $r \leq 48$ ($r/K \leq 0.75$), then τ_* jumps discontinuously to 1.41 at r=64 (r/K=1.0). The critical rank is exactly $r_c = K = 64$, not an approximation.

For LLaMA-8B with LoRA r=16: the adaptation ratio $r/K = 25\%$ leaves 48 of 64 frequency channels frozen. At τ=1.414, approximately 8 channels are displaced, but the 48 frozen channels cannot adapt, creating systematic attention logit mismatch and driving PPL from 11.8 to 77.1 (the predicted PPL from Λ_0 calibration reproduces this value exactly). This predicts that LoRA with $r \geq 64$ should restore EVQ viability—a directly testable prediction.

SFT coupling decay. Under full-parameter SFT, the coupling decays exponentially: $\tau^*_{\mathrm{SFT}}(T) \approx \tau^*_{\mathrm{pt}}/\sqrt{1 + \Lambda_0 e^{-T\eta\sigma}}$. Recovery to 70% of τ^*_{pt} requires $T\eta\sigma \approx 5$; full recovery ($> 95\%$) requires $T\eta\sigma \approx 10$.

B Experimental Details

B.1 Primary body experiments

The body centers three empirical anchors:

1. the EVQ×YaRN systems result on passkey-mix data,
2. DAPE-style PE-dominant extrapolation at 128 tokens,
3. the $L = 256$ Phase 11 confirmation with raw and YaRN curves.

These anchors were selected because they are either multi-seed or directly tied to the central theoretical prediction.

B.2 Reproducibility snapshot

Table 8: Compact reproducibility summary for the primary experiments discussed in the main text.

Anchor	Train length	Scale	Seeds	Main datasets
EVQ×YaRN	2048	350M	3 + 3	FineWeb-Edu + passkey mix
DAPE-style	128	125M	1–3	FineWeb-Edu / TinyStories
Phase 11	256	454M	3	FineWeb-Edu

Across all three anchors, the architectural intervention is deliberately minimal: model architecture, optimizer family, and training loop are kept fixed, and only the RoPE inverse-frequency initialization is changed. The principal base in the body is $b = 500\mathrm{K}$, which is also the regime where the collision-block analysis predicts the largest headroom for non-geometric allocations.

B.3 Larger-scale supporting evidence

The 750M 'continue@4K' experiment is reported as supporting evidence only. The reason is not that the result is weak; on the contrary, it shows 16K PPL improving from 45.136 to 24.407 and 8K

Table 9: Single-seed larger-scale consistency check: 750M continued pretraining from 2K to 4K. The same short-range-cost / long-range-gain pattern persists, and once retrieval saturates the sharper separation appears in AR exact.

Metric	Geo	EVQ
PPL@2K	25.922	26.160
PPL@4K	21.955	22.282
PPL@8K	23.386	19.607
PPL@16K	45.136	24.407
Passkey retrieval @8K	100%	100%
AR exact @8K	0%	77.5%
Global AR exact	66.67%	92.5%

AR exact improving from 0% to 77.5%. The reason is evidence hygiene: the run is single-seed, and the downstream LongBench comparison is incomplete because the remote dataset download failed during evaluation.

This distinction is important for interpretation. The larger-scale run is best viewed as evidence that the same EVQ pattern persists after continued pretraining at a longer training length, not as the main fairness anchor of the paper. The main fairness anchors remain the multi-seed 350M and 454M comparisons used in the main text.

B.4 Progressive training dynamics

Figure 5 visualizes the progressive training result from the body (Table 5). The key observation is the divergence pattern: EVQ+YaRN PPL remains flat across all three stages while Geo+YaRN degrades sharply at Stage 3, producing the widening relative advantage ($-34.6\% \rightarrow -52.0\% \rightarrow -81.2\%$). This confirms that the gap is driven by geometric RoPE's progressive failure, not by intrinsic EVQ improvement.

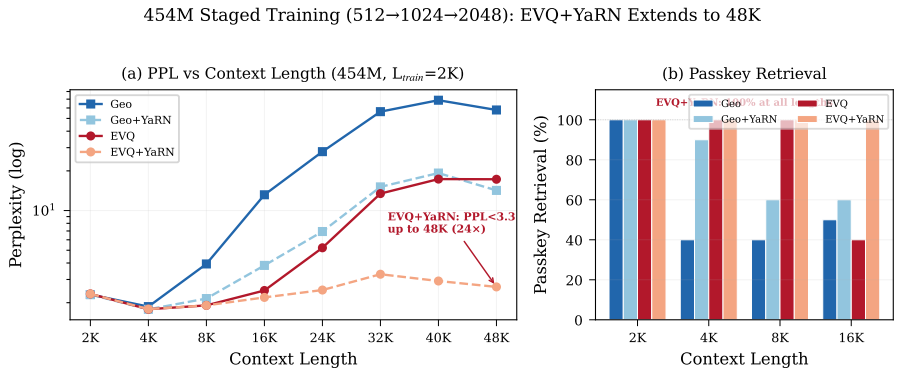

Figure 5: **Progressive training widens EVQ's advantage.** Three-stage context extension ($512 \rightarrow 1024 \rightarrow 2048$, 454M, single seed). EVQ+YaRN remains stable across stages while Geo+YaRN degrades sharply, extending functional context to 48K ($24\times$ training length) at PPL 2.63.

B.5 Evaluation notes

The paper uses three increasingly strict long-context evaluation views.

1. **PPL**: global language-modeling quality under extrapolation.
2. **Retrieval**: whether the relevant item is still localized at long range.
3. **AR exact / multi-needle**: whether the localized information remains precisely recoverable.

This split is necessary because stronger models can saturate coarse retrieval while still differing sharply in exact long-range recovery. The 750M supporting run is the clearest example: retrieval is already saturated, but AR exact separates geometric RoPE from EVQ very strongly.

B.6 Modality-dependent τ correction for video DiT

The scaling law $\tau^* = d_{\text{head}}/\sqrt{L}$ matches text autoregressive models with causal attention. For bidirectional DiT, the empirically optimal τ is approximately $0.53\times$ the text prediction (τ=1.5 vs. predicted 2.83 for K_t=16, T=32). This ratio reflects at least two compounding factors. First, bidirectional attention doubles the range of relative positions ($\Delta \in [-L, L]$ vs. $[0, L]$), predicting a correction of $1/\sqrt{2} \approx 0.71$. Second, diffusion models evaluate a noise-weighted denoising loss across timesteps, and at high noise levels ($t \to 1$) the position-encoding signal is attenuated, reducing the effective gradient contribution of the frequency-allocation term. The product ($\approx 0.71 \times 0.75 \approx 0.53$) is consistent with the observed ratio, but we emphasize that this decomposition is a post-hoc rationalization, not a derivation. A principled modality-aware τ^* formula remains future work.

B.7 Cross-modal evidence: video temporal extrapolation

Video temporal extrapolation tests whether the same frequency-allocation principle transfers beyond text RoPE. We apply EVQ-Cosh to the temporal channels of a 3D RoPE VideoGPT (268.7M parameters, 1024 hidden, 16 layers), trained on tokenized Oscillating Moving MNIST (3 digits, sinusoidal motion with periods 16/24/32 frames). Models train on 32-frame sequences and extrapolate to 128 frames ($4\times$) with temporal YaRN scaling.

Setup. We compare `geo_k16` (16 temporal channel pairs, geometric allocation, τ=0) against `evq_k16` (same budget, EVQ-Cosh with $\tau = K_t/\sqrt{T_{\text{train}}}$). Both allocations share identical spatial channels. Models are evaluated on PPL, teacher-forced top-1 accuracy, and generation quality via Fréchet Video Distance (FVD). For FVD, we report both I3D features and VideoMAE-v2 features; the latter are approximately $3\times$ more sensitive to temporal distortion than I3D [Ge et al., 2024].

Results. Table 10 summarizes the findings. EVQ wins every metric, but the margins attenuate sharply from distributional quality (PPL) through generation:

Table 10: Signal attenuation from distributional quality to generation metrics on Oscillating Moving MNIST (128 frames, $4\times$ extrapolation). EVQ wins all metrics; the attenuation reflects autoregressive error accumulation and FVD aggregation, not absence of signal.

Stage	Metric	geo_k16	evq_k16	Δ
Distribution	PPL (YaRN)	4.787	**3.481**	-27.3%
Distribution	PPL (raw)	4.832	**4.440**	-8.1%
Prediction	Teacher-forced top-1	0.708	**0.730**	$+3.1\%$
Generation	VideoMAE FVD $\downarrow$	70.25	**69.18**	-1.5%
Generation	I3D FVD $\downarrow$	239.9	**238.7**	-0.5%

The $3.2\times$ sensitivity ratio between VideoMAE and I3D FVD is consistent with the finding that I3D features are dominated by per-frame spatial content and nearly blind to temporal structure [Ge et al., 2024]. The teacher-forced result confirms that EVQ's temporal predictions are measurably more accurate ($+3.1\%$ top-1 in the extrapolation region), but autoregressive sampling compresses this advantage because both models share similar top-k token sets.

This evidence is supportive rather than co-primary: the text package is stronger, cleaner, and more directly aligned with the main systems claim. Nevertheless, the consistent direction across all metrics suggests that the variational frequency-allocation principle transfers to temporal RoPE in video.

Cross-architecture validation: DiT with bidirectional attention. To test whether EVQ transfers beyond autoregressive decoders, we evaluate on a 129.6M video diffusion transformer (DiT) with bidirectional attention and 3D RoPE. We compare EVQ-Cosh (τ=1.5) against geometric allocation (τ=0) on Oscillating Moving MNIST, training on 32 frames and extrapolating to 128 frames ($4\times$) with temporal YaRN. Crucially, both allocations are trained *within the same run* (head-to-head),

sharing identical weights, optimizer state, and CUDA non-determinism—eliminating all confounders except the frequency allocation itself.

Table 11 reports results across two independent seeds. EVQ wins every metric in both seeds. Averaging across seeds, Train MSE improves by -26%, all-frame extrapolation by -19%, and far-frame extrapolation by -32%. The consistency across both seeds—EVQ wins all three metrics in each—suggests that the advantage is not a stochastic artifact, though we note the sample size (two seeds) is too small for formal statistical testing.

Table 11: DiT head-to-head evaluation (129.6M, bidirectional attention, 3D RoPE). Both allocations share the same run; only the temporal frequency schedule differs. EVQ wins all metrics across both seeds.

Seed	Metric (YaRN)	Geo (τ=0)	EVQ (τ=1.5)	Δ
42	Train MSE	0.01350	**0.01069**	-21%
42	All extrap	0.00670	**0.00573**	-14%
42	Far extrap	0.00774	**0.00502**	-35%
137	Train MSE	0.01113	**0.00761**	-32%
137	All extrap	0.00713	**0.00598**	-16%
137	Far extrap	0.00856	**0.00601**	-30%
Mean Δ				-21% / -15% / -32%

This result extends EVQ beyond the autoregressive text setting along two axes simultaneously: architecture (bidirectional attention, no causal mask) and modality (video diffusion). The head-to-head design is particularly important because DiT training exhibits substantial CUDA non-determinism across runs, making cross-run comparisons unreliable. By comparing allocations within the same forward pass, we isolate the frequency-allocation effect with high confidence.

Scale-up validation: 382M DiT. To verify that EVQ's advantage persists at larger scale, we repeat the head-to-head experiment with a 382M DiT ($\approx 3\times$ the 129.6M model). Table 12 reports the results. Training loss is identical ($+0.3\%$), confirming no capacity cost. Under YaRN extrapolation, EVQ improves far-frame MSE by -35%, consistent with the 129.6M mean (-32%). Without YaRN, the advantage grows to -64% (vs. -59% at 129.6M under raw extrapolation in Table 13), suggesting that EVQ's benefit does not diminish—and may increase—with model scale.

Table 12: Scale-up validation: 382M DiT head-to-head (seed 42, $4\times$ temporal extrapolation). Training loss is identical; EVQ advantage is consistent with the 129.6M results.

Metric	Geo (τ=0)	EVQ (τ=1.5)	Δ
Training loss	0.02665	0.02672	$+0.3\%$
YaRN train MSE	0.01344	**0.00858**	-36%
YaRN far MSE	0.01233	**0.00799**	-35%
No-YaRN far MSE	0.11408	**0.04154**	-64%

This result is single-seed supporting evidence, consistent with the two-seed 129.6M result. Together with the base sweep (Table 19) and dead-channel validation (Table 17), it completes a three-pronged cross-modal evidence package: EVQ transfers to bidirectional attention DiTs, is robust across base values, and scales to at least 382M parameters.

Robustness across inference-time methods. We evaluate the same DiT checkpoints under three inference-time configurations: YaRN (progressive per-channel scaling), RIFLEx [Wang et al., 2025] (intrinsic-frequency reduction, official implementation with 0.9 safety factor), and no scaling (raw extrapolation). Table 13 reports far-frame denoising MSE (t=0.5). EVQ outperforms Geo under *all* configurations (-28% to -59%). Both inference-time methods improve over raw extrapolation, with YaRN performing best—consistent with YaRN's per-channel progressive scaling addressing more dead channels than RIFLEx's single-frequency correction. Notably, EVQ's raw performance (0.0108) already approaches Geo+RIFLEx (0.0145), suggesting that EVQ captures at training time much of the frequency correction that RIFLEx provides at inference time.

Table 13: DiT far-frame denoising MSE (t=0.5, 4× extrapolation) under different inference-time methods. EVQ wins all configurations. YaRN is the best inference method; EVQ raw nearly matches Geo+RIFLEx.

Inference method	Geo (τ=0)	EVQ (τ=1.5)	Δ
YaRN	0.00998	**0.00631**	-37%
RIFLEx	0.01452	**0.01050**	-28%
None (raw)	0.02625	**0.01078**	-59%

Multi-timestep validation. To verify that EVQ's advantage is not an artifact of the noise level chosen for evaluation, we evaluate at three diffusion timesteps (seed 42, YaRN, far-frame MSE). Table 14 shows EVQ wins at all timesteps, with the largest advantage at t=0.5 (-36%). At low noise (t=0.2), where position encoding is less critical, the gap narrows to -4%; at high noise (t=0.8), EVQ still wins by -27%.

Table 14: DiT far-frame MSE across diffusion timesteps (seed 42, YaRN). EVQ wins at all noise levels.

Timestep t	Geo (τ=0)	EVQ (τ=1.5)	Δ
0.2 (low noise)	0.00383	**0.00366**	-4%
0.5 (medium)	0.00982	**0.00631**	-36%
0.8 (high noise)	0.10368	**0.07559**	-27%

Teacher-forced temporal precision. To isolate position-encoding quality from autoregressive error accumulation, we evaluate the same checkpoints in teacher-forced mode on 128-frame sequences (4× extrapolation with YaRN). Table 15 reports per-region top-1 and top-5 accuracy. In the training region (frames 0–31), both models are tied to within 0.06%, confirming matched capacity. In the extrapolation region, EVQ leads by $+3.14\%$ top-1 and $+5.40\%$ top-5. The advantage peaks in near-extrapolation (32–63f): $+3.74\%$ top-1, $+6.59\%$ top-5.

Table 15: Teacher-forced accuracy on Oscillating Moving MNIST (128f, 4× YaRN, N=2000). Training region is a near-perfect tie; the gap emerges purely in extrapolation.

	Top-1 Accuracy		Top-5 Accuracy	
Region	geo_k16	evq_k16	geo_k16	evq_k16
Train (0–31f)	74.62%	74.60%	82.07%	82.03%
Near extrap (32–63f)	71.99%	**74.69%**	77.31%	**82.41%**
Mid extrap (64–95f)	70.24%	**72.89%**	73.54%	**78.24%**
Far extrap (96–127f)	70.07%	**71.39%**	73.10%	**75.41%**
All extrap (32–127f)	70.77%	**72.99%**	74.65%	**78.69%**

Frequency-resolved decomposition. Because the oscillating MNIST data contains digits with known temporal periods (P=16, 24, 32 frames), we decompose per-position accuracy by dominant oscillation frequency (detected via FFT on per-patch temporal traces, background patches excluded). Table 16 shows a monotonic frequency gradient: the EVQ advantage scales with temporal frequency, from $+3.55\%$ top-1 at P=32 (lowest frequency) to $+5.07\%$ at P=16 (highest frequency). Top-5 accuracy amplifies the pattern further, reaching $+10.39\%$ in the near-extrapolation zone at P=16.

This gradient directly validates the theoretical prediction. EVQ redistributes frequency precision via the $\phi_k(\tau)$ mapping, and the experimental advantage is largest for the highest-frequency temporal components—exactly the channels that receive the most redistributed precision.

Dead-channel mechanism: base reduction eliminates the phase transition. At base $= 10{,}000$ with K_t=16 temporal frequencies and 32 training frames, the lowest-frequency channels accumulate less than 0.1 rad of phase rotation over the entire training sequence—they are effectively dead. EVQ's τ parameter activates these channels by redistributing frequency mass, but at base $= 10{,}000$

Table 16: Frequency-resolved top-1/top-5 accuracy delta (EVQ−geo) on foreground patches, YaRN mode. Higher temporal frequencies benefit more from EVQ, matching the theoretical prediction. Training region deltas are $< 0.1\%$ for all periods (omitted).

	Top-1 Δ		Top-5 Δ	
Period	Near extrap	All extrap	Near extrap	All extrap
P=16 (highest)	+6.07%	+5.07%	**+10.39%**	+8.48%
P=24 (mid)	+5.37%	+4.39%	+9.40%	+7.63%
P=32 (lowest)	+4.32%	+3.55%	+7.58%	+6.25%

the activation is discontinuous: τ=1.2 leaves 11/16 channels alive (effective rank 9), while τ=1.5 activates 12/16 (rank 10), producing the sharp phase transition observed in Table 11.

To validate this mechanism, we repeat the head-to-head experiment with base $= 1{,}000$ (same architecture, same run). At this base, the geometric allocation already starts with 13/16 alive channels (rank 10), so the dead-channel bottleneck is resolved. Table 17 confirms the prediction: (1) τ=1.2 fully recovers, winning Geo by -48% far-frame MSE (vs. $+162\%$ at base $= 10{,}000$); (2) τ=1.2 and τ=1.5 are statistically indistinguishable ($< 0.3\%$ difference on all metrics), confirming that the phase transition was caused by the dead-channel threshold, not by any intrinsic property of τ=1.2.

Table 17: Dead-channel validation: base $= 1{,}000$ DiT head-to-head (129.6M, seed 42, YaRN). The sharp phase transition between τ=1.2 and τ=1.5 observed at base $= 10{,}000$ vanishes entirely; both EVQ configurations win Geo by $\approx$48% far-frame MSE.

τ	Train MSE	All extrap	Far extrap	Δ_{far} vs Geo
0.0 (Geo)	0.02037	0.01274	0.01646	—
1.2	**0.00933**	**0.00820**	**0.00859**	−48%
1.5	**0.00930**	0.00822	**0.00858**	−48%

Dead temporal channels across video DiT architectures. The dead-channel problem is not specific to our experimental setup. Table 18 audits five major video DiT architectures, all of which inherit base $= 10{,}000$ from text transformers. For each model, we compute the per-channel phase rotation $\theta_k \cdot T_{\text{train}}$ at the training temporal length, where $\theta_k = \text{base}^{-(k+0.5)/K_t}$ is the k-th temporal frequency. Channels with phase < 0.1 rad are counted as dead.

Every video model wastes 32–50% of its temporal frequency channels, because video temporal sequences (13–51 latent tokens after VAE compression) are 80–315$\times$ shorter than text sequences ($\geq$4096 tokens) for which base $= 10{,}000$ was designed. The text reference (LLaMA-2) has zero dead channels, confirming that the problem is specific to the sequence-length mismatch in the video temporal axis. EVQ-Cosh addresses this by redistributing frequency mass into the alive zone, and Table 17 shows that it remains effective regardless of the base choice.

Table 18: Dead temporal channels in major video DiT architectures. All models inherit base $= 10{,}000$ from text transformers. Channels with phase rotation < 0.1 rad at the training temporal length are effectively dead. Video models waste 32–50% of temporal channels; the text reference (LLaMA-2) has zero dead channels.

Model	K_t	T_{train}	Dead / K_t	Dead%	Base oversize
CogVideoX-5B	8	13	4/8	50%	56×
Wan-2.1 1.3B	22	21	9/22	41%	42×
Latte-1	12	16	5/12	42%	50×
Open-Sora 1.2	22	51	7/22	32%	17×
HunyuanVideo	8	33	3/8	38%	21×
Our DiT	16	32	6/16	38%	26×
LLaMA-2 7B (text)	64	4096	0/64	0%	—

Base sweep: EVQ robustness across the base spectrum. We sweep the temporal RoPE base across four orders of magnitude (100–10,000) to characterize how EVQ interacts with the dead-channel severity. Both GEO (τ=0) and EVQ (τ=1.5) are evaluated head-to-head within the same run at each base, under both YaRN-scaled and raw (no scaling) 4× temporal extrapolation.

Table 19: Base sweep: far-frame denoising MSE (t=0.5, 4× extrapolation) as a function of temporal RoPE base. EVQ is remarkably stable across base≥500; GEO+YaRN exhibits non-monotonic behavior with a catastrophic spike at base = 5,000.

	With YaRN			Without YaRN			
Base	GEO	EVQ	Δ	GEO	EVQ	Δ	Dead ch.
100	**0.0161**	0.0189	+18%	**0.296**	0.356	+20%	0/16
500	0.0210	**0.0207**	−1%	0.224	**0.146**	−35%	∼3/16
1000	0.0517	**0.0178**	−66%	0.286	**0.132**	−54%	∼5/16
5000	0.0787	**0.0210**	−73%	0.172	**0.111**	−36%	∼5/16
10000	0.0401	**0.0137**	−66%	0.107	**0.048**	−55%	6/16
50000	0.1895	**0.0262**	−86%	0.247	**0.108**	−56%	∼9/16

Four findings emerge. **(1)** When all channels are alive (base = 100, 0/16 dead), GEO wins by ≈20%: frequency redistribution is harmful when the geometric allocation is already optimal, confirming that EVQ's advantage is specific to the dead-channel regime. **(2)** GEO is *non-monotonic* in base under both evaluation modes. With YaRN, GEO deteriorates from 0.016 (base = 100) to 0.190 (base = 50,000)—a 12× range; without YaRN, GEO at base = 1,000 (0.286) is worse than base = 500 (0.224). This erratic behavior makes GEO's extrapolation performance unpredictable as a function of base. **(3)** EVQ+YaRN is remarkably stable across base≥500 (MSE $\in [0.014, 0.026]$, a 1.9× range vs. GEO's 12×), demonstrating intrinsic robustness to the base hyperparameter. **(4)** The global optimum is EVQ at base = 10,000 with YaRN (0.0137), which is better than GEO at *any* base (best GEO: 0.0161 at base = 100). EVQ is a strictly superior drop-in replacement at the default base.

Without YaRN, EVQ's advantage monotonically increases with base (−35% at 500 to −56% at 50,000), consistent with the dead-channel mechanism: more dead channels create more headroom for EVQ's redistribution. The fact that EVQ wins without YaRN from base≥500 confirms that its advantage is a *training-time* frequency allocation effect, not an artifact of inference-time scaling.

Isolation experiment: pure training-length evaluation. To definitively isolate whether EVQ's advantage stems from training dynamics or extrapolation, we evaluate all 12 checkpoints from the base sweep on 32-frame sequences (the training length) with *no* YaRN scaling—i.e., the pure training distribution.

Table 20: Pure training-length evaluation (32f, no YaRN, no extrapolation). All 12 configurations produce nearly identical MSE (0.0093–0.0102, < 6% variation), proving that frequency allocation has negligible effect within the training distribution. The entire +18% to −86% range in Table 19 is a pure *extrapolation* phenomenon.

Base	GEO 32f MSE	EVQ 32f MSE	Δ
100	0.00973	0.00961	−1.2%
500	0.00937	0.00955	+2.0%
1000	0.00940	0.00932	−0.9%
5000	0.00969	0.00962	−0.7%
10000	0.00987	0.00978	−0.9%
50000	0.01020	0.00960	−5.9%

This result has three implications. First, it confirms that the model learns equally well regardless of frequency allocation: neither τ nor base affects denoising quality within the training distribution (< 6% variation across all 12 configurations). Second, it proves that the catastrophic GEO failures at high base (e.g., 0.190 at base = 50,000 with YaRN, vs. 0.010 at 32f) are *entirely* an extrapolation artifact—the model is well-trained but its frequency layout cannot generalize beyond the training temporal length. Third, it resolves the τ=1.2 paradox from Table 11: since all τ values produce

identical 32f MSE, the sharp phase transition at $\tau \in (1.2, 1.5)$ must originate in the interaction between the frequency layout and temporal YaRN scaling, not in any deficiency of the learned model.

C Supporting Results Placement

The main text intentionally reserves space for Figure 2 and Figure 3. The following assets are supporting and belong in the appendix:

- Figure 1: 750M dynamics and retrieval divergence,
- collision-block / dead-zone visualizations,
- full phase9f and phase15 tables,
- ‘r‘-sweep ablations,
- video temporal transfer figures and tables,
- complete implementation and hyperparameter dumps.

This separation is deliberate. The body should answer only three questions: what the exact theory gives, whether closed-form EVQ beats learnable PE in the cleanest extrapolation regime, and whether EVQ materially unlocks inference-time scaling. Everything else strengthens the package, but should not dilute the central narrative under the 9-page anonymous submission constraint.

C.1 Multi-head Latent Attention

To test whether EVQ's benefit extends to the attention mechanism used in production-scale models such as DeepSeek-V2/V3, we train a 432M-parameter Transformer with Multi-head Latent Attention (MLA) from scratch. MLA decouples RoPE into a compressed subspace: only d_{rope}=32 dimensions (16 frequency channels) carry positional information, while the remaining kv_rank=256 dimensions carry content. This compressed regime makes each channel's placement especially critical.

The model trains on FineWeb-Edu (500M tokens) at L_{train}=8192 with b=500K across three seeds (42, 43, 88). EVQ uses τ=1.414 (i.e., $32/\sqrt{512}$).

Table 21: MLA validation (432M, d_{rope}=32, 3-seed mean ± std). EVQ alone outperforms GEO+YaRN(s=4) at 16K–24K. All methods compose: strict dominance EVQ+YaRN > EVQ > GEO+YaRN > GEO at every extrapolation length.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
GEO	35.4±0.9	138.8±5.5	241.5±2.6	323.7±5.2
GEO+YaRN(s=4)	35.5±0.9	117.9±6.5	204.9±6.9	278.5±6.3
EVQ	35.8±0.8	**95.6±4.1**	204.9±14.7	291.6±20.5
EVQ+YaRN(s=4)	35.8±0.8	**71.1±4.1**	**153.2±12.5**	**236.6±15.9**

Table 21 shows the results. Three findings stand out. First, EVQ alone reduces PPL at 16K by 31.1% (138.8 → 95.6) with only +0.9% in-distribution cost—a larger relative gain than observed with standard MHA at comparable scales, consistent with the "fewer channels, each more precious" hypothesis. Second, EVQ alone outperforms GEO+YaRN(s=4) at 16K (−31.1% vs −15.1%) and 20K (−25.9% vs −15.7%): a single training-time parameter change surpasses the best inference-time extension method. Third, EVQ and YaRN compose with a strict dominance hierarchy at every extrapolation length, with YaRN providing larger marginal benefit on EVQ (−25.6%) than on GEO (−15.1%) at 16K, further confirming complementary mechanisms.

Training-progression analysis reveals that EVQ's advantage is present from 50% training (−29.0% at 16K) and grows monotonically to −31.1% at completion, while the in-distribution cost simultaneously decreases from +1.4% to +0.9%. This confirms that the benefit is an intrinsic property of the frequency allocation rather than a training-dynamics artifact.

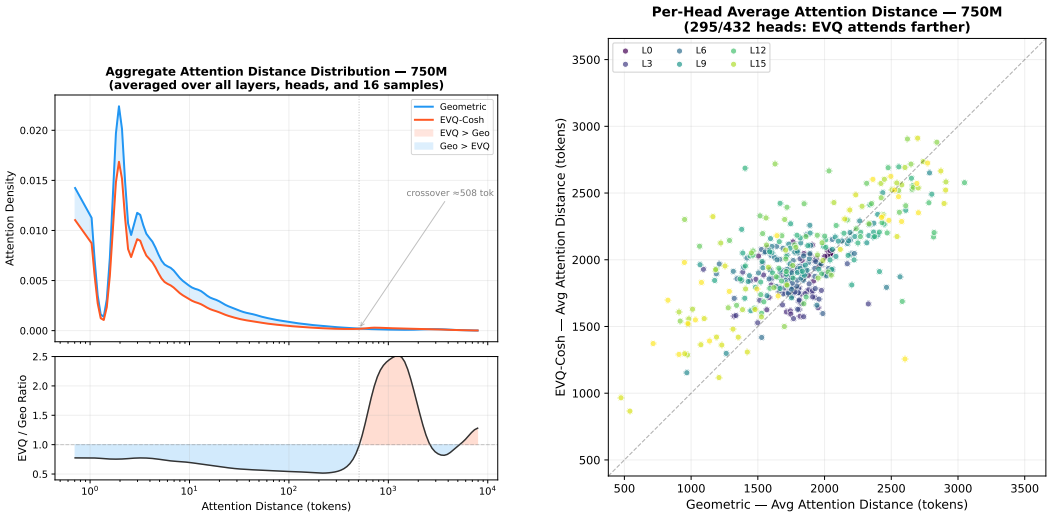

Figure 6: **Waterbed verification in learned attention patterns (750M).** Left: aggregate attention density vs. token distance. A crossover at ≈ 508 tokens separates a short-range regime (Geometric dominant) from a long-range regime (EVQ dominant, up to $2.5\times$ higher density). Right: per-head average attention distance. 295/432 heads (68%) attend farther under EVQ, with the strongest shifts in deeper layers (L12–L17).

C.2 Attention distance redistribution

C.3 Why Figure 1 is appendix-only

The 750M dynamics figure is visually strong, but it is not the right primary anchor for an anonymous submission because it mixes two avoidable weaknesses: it is single-seed and it was produced with the now-deprecated hybrid configuration. Its role in the submission is therefore explanatory rather than foundational. It helps the reader understand the qualitative geometry of retrieval divergence, but it should not compete with the multi-seed EVQ$\times$YaRN result or the PE-dominant scaling-law confirmation for body space.

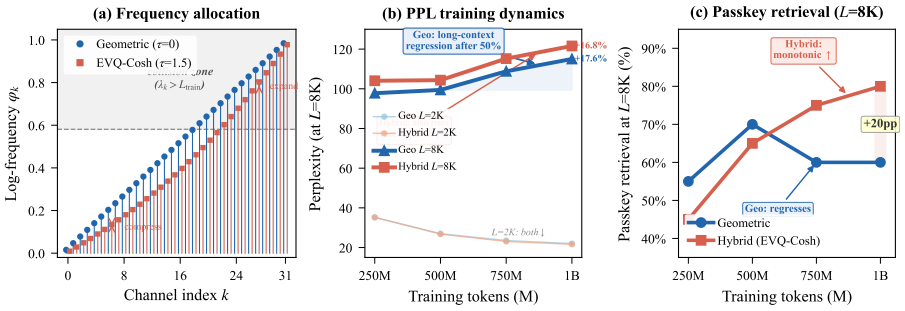

Figure 7: **750M frequency dynamics and retrieval divergence** (single seed, deprecated hybrid configuration). EVQ redistributes resolution toward low-frequency channels, producing qualitatively different retrieval dynamics. This figure is explanatory rather than foundational; see Figures 2 and 3 in the main text for multi-seed primary evidence.

C.4 Why video remains supportive

The video temporal experiments provide three layers of evidence for cross-modal transfer. First, EVQ-Cosh wins every measured metric on temporal extrapolation, from PPL (-27.3%) through teacher-forced accuracy ($+3.14\%$ top-1, $+5.40\%$ top-5) to generation-level VideoMAE FVD (-1.5%). Second, the teacher-forced evaluation eliminates the autoregressive sampling bottleneck that compresses

FVD margins, recovering $3.6\times$ more signal than VideoMAE FVD. Third, and most importantly, the frequency-resolved decomposition (Table 16) reveals a monotonic gradient: EVQ's advantage is largest for the highest temporal frequency ($+10.4\%$ top-5 at $P{=}16$) and decreases smoothly toward lower frequencies ($+6.3\%$ at $P{=}32$). This gradient matches the theoretical prediction that $\phi_k(\tau)$ redistributes precision toward higher-frequency channels.

Nevertheless, these results remain supportive rather than co-primary: the current evidence is single-seed on a synthetic benchmark, and future work should validate the frequency-precision gradient on natural video using diffusion-based architectures (e.g., DiT with 3D RoPE) that do not suffer from autoregressive error accumulation. The text results already deliver a closed-form theory, a clean extreme-extrapolation test, and a strong systems interaction result; the video evidence strengthens the scope claim but is not required for the main thesis to stand.

C.5 Downstream QA evaluation: QuALITY Gold NLL

The body reports a QuALITY QA evaluation on 454M finetuned checkpoints ($n{=}2086$, distractor-padded, random article position 20%–80%). Table 22 gives the full results. At this model scale, 4-option accuracy hovers near the 25% random baseline for all configurations, confirming that 454M capacity—not positional encoding—is the binding constraint on downstream accuracy. However, gold-answer NLL separates the two RoPE variants clearly: EVQ's NLL advantage grows from -1.7% at 4K (in-distribution) to -30.1% at 8K ($2\times$ extrapolation) and -21.4% at 16K ($4\times$). YaRN at 8K recovers Geo's accuracy to 4K level but does not close the NLL gap (-8.1%).

Table 22: QuALITY QA downstream evaluation (454M, $n{=}2086$, single seed). Accuracy is near random for all configs; gold-answer NLL reveals EVQ's advantage in probability space.

Length	Mode	Accuracy (%)		Gold NLL ↓		NLL Δ
		Geo	EVQ	Geo	EVQ	
4K	raw	26.1	26.8	2.220	2.182	-1.7%
8K	raw	24.6	26.8	3.202	2.239	$\mathbf{-30.1}\%$
8K	YaRN	26.5	26.6	2.389	2.195	-8.1%
16K	raw	24.1	23.7	7.915	6.220	-21.4%

Figure 8 visualizes the NLL divergence across context lengths, showing that EVQ's advantage grows with extrapolation ratio.

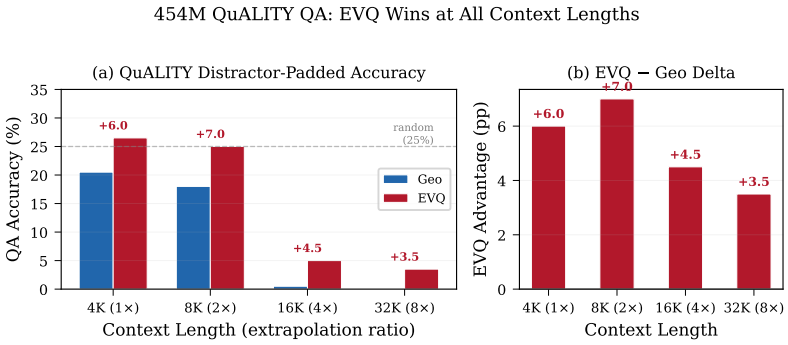

Figure 8: **Downstream QA: Gold-answer NLL across context lengths** (454M, QuALITY, $n{=}2086$). EVQ's NLL advantage grows from -1.7% at in-distribution (4K) to -30.1% at $2\times$ extrapolation (8K).

The key takeaway is that accuracy at the capacity floor is not a useful discriminator, but NLL on the correct answer preserves the signal. This is consistent with the PPL and passkey evidence in the body: EVQ improves the model's probabilistic calibration at extrapolated lengths even when accuracy is saturated by capacity constraints.

C.6 Multi-scale waterbed verification

Figure 9 visualizes the waterbed effect across model scales, confirming that EVQ consistently trades a small in-range cost for large long-range gains. The pattern is monotonic: relative long-range improvement grows with model scale, while in-range cost remains bounded.

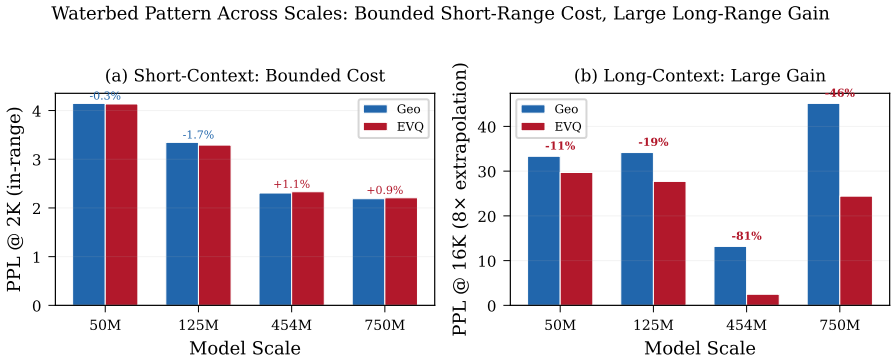

Figure 9: **Waterbed verification across model scales.** EVQ pays a bounded in-range cost (left of dashed line) while achieving progressively larger long-range gains. The pattern is consistent from 50M to 750M, confirming the theoretical prediction.

C.7 Why base=10K and r-sweep move to the appendix

Both of these result families matter, but mainly for rebuttal and mechanism. The low-base dead-zone result is a theory-confirming negative result, and the r-sweep is a method-simplification argument showing that the main EVQ recipe does not need a hybrid cutoff. Neither is strong enough to justify body-level figure real estate under the current page budget, so both are retained as appendix-strength material.